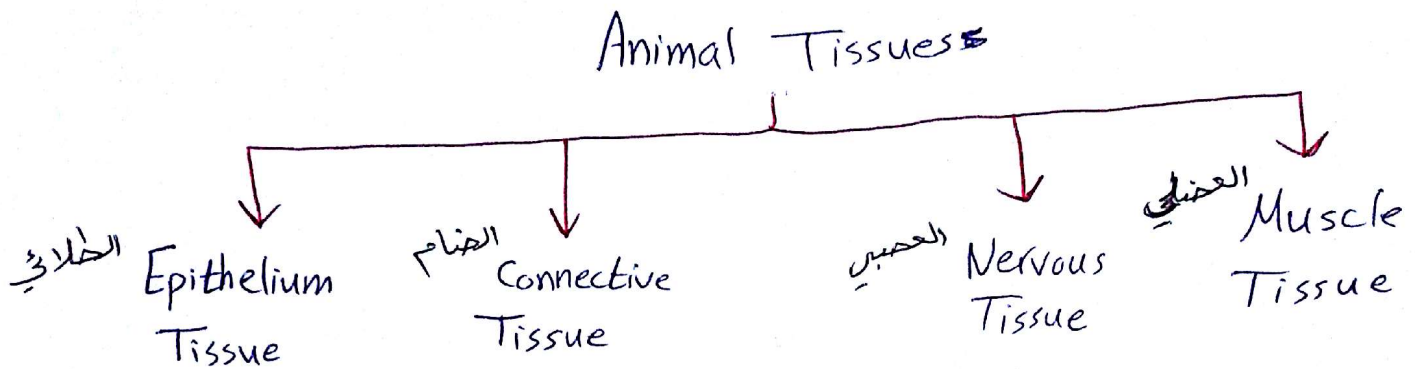


Lab # 11 :- Animal Tissues

- * Tissue → a cellular organizational level intermediate between cells and a complete organs.
- * Histology → the study of animal tissue.



1* Epithelium :-

- is from the **endothelium** → a specialized type of epithelium that composes the vasculature.
- A True epithelial tissue is present only in a single layer in a single layer of cells held together via tight junctions.
- Covers all organismal surfaces that come in contact with the external environment such as: **skin**, airways, **digestive tract**.
- Functions → protection, secretion, absorption.
- This tissue is ~~is~~ separated from other below tissues by a basal lamina (basement membrane).

- In general, epithelium tissues are classified by:-

* the morphology of their cells.

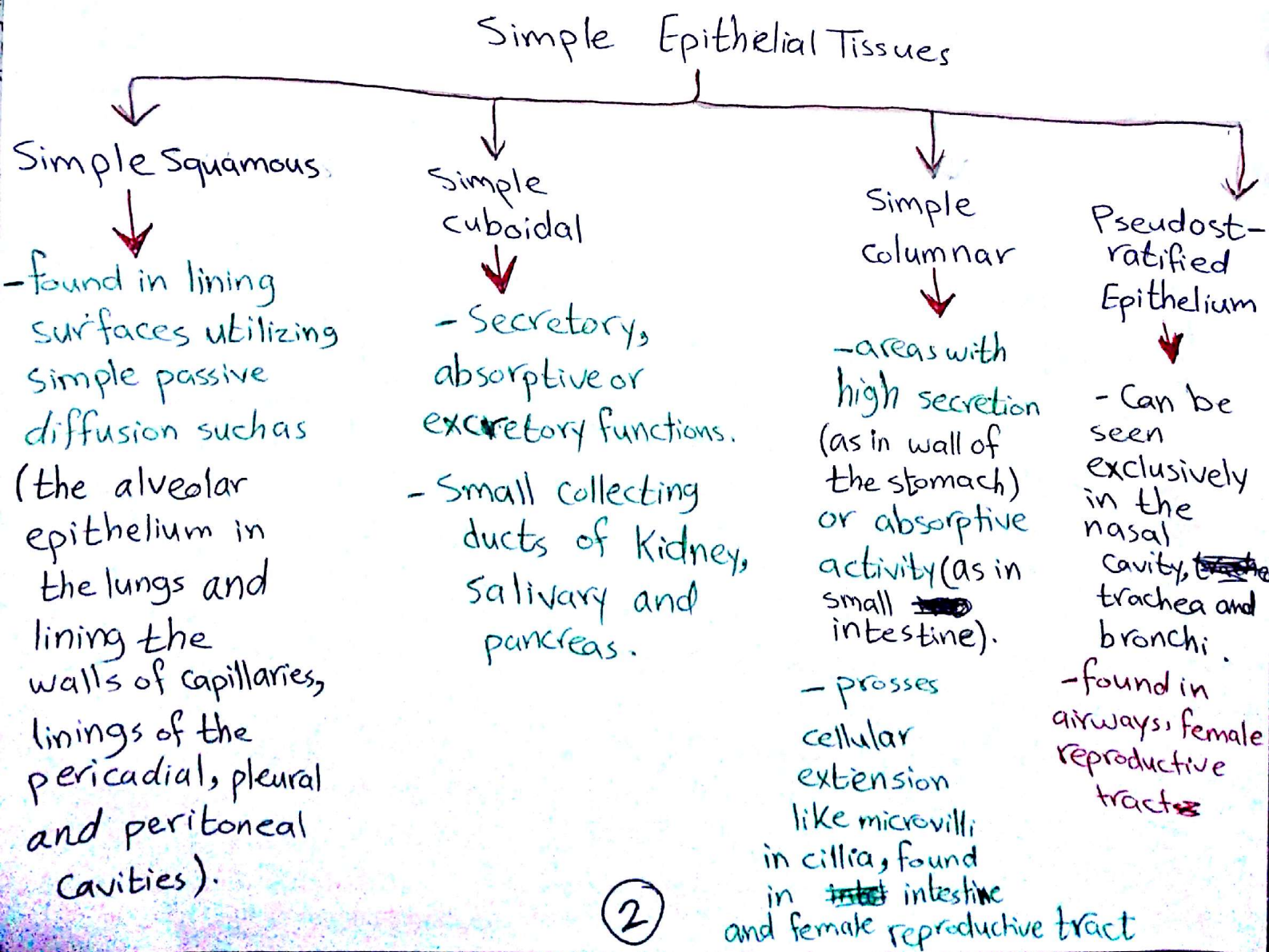
* the number of their layers.

• Simple epithelium → one layer of cells in contact with the underlying basement membrane.

• Stratified epithelium → made up from two or more layers.

* Simple Epithelial Tissues:-

- are classified by the shape of their cells.



* Stratified Epithelium:-

- Multilayered tissue found where body linings have to withstand mechanical or chemical insult, it can be abraded and lost without exposing sub epithelial layers.
- Cells become more flatten as the layers become more apical, though in their most basal layers the cell ~~can~~ can be:-

Stratified Epithelium

Stratified Squamous

- function (protection)
- found in variety lines wet surfaces like (lining of the mouth, esophagus, part of epiglottis, vagina and covers the tongue).
- Keratinized cells are ~~dead~~ dead and lose nucleus and cytoplasm, instead contain a ~~thick~~ tough, resistant protein called Keratin.

Stratified Cuboidal

- Two or more layers of cells which cells in the apical are ~~cube~~ cube-shaped
- functions in protection, secretion and absorption.
- located in ducts of adult sweat and esophageal glands, and parts of male urethra.

Stratified Columnar

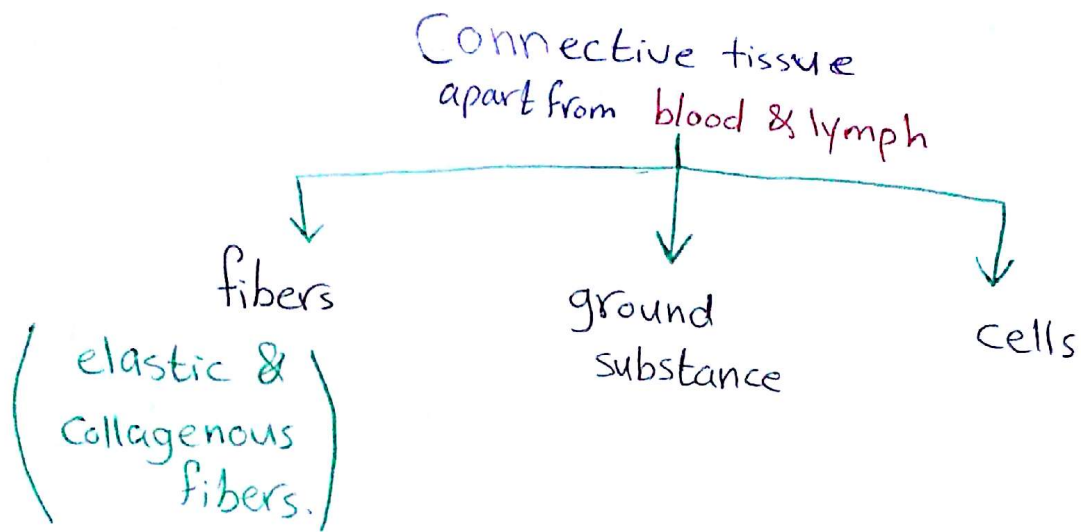
- functions in protection and secretion.
- rare in the body, small amounts in male urethra and large ~~are~~ ducts of some glands

Transitional epithelium

- found in tissues that stretch.
- it can appear to be ~~stratified~~ stratified cuboidal when the tissue is not stretched or stratified squamous when the organ is distended.
- found in the bladder, ureters, and urethra. ~~الكلى~~ ^{الكلى}

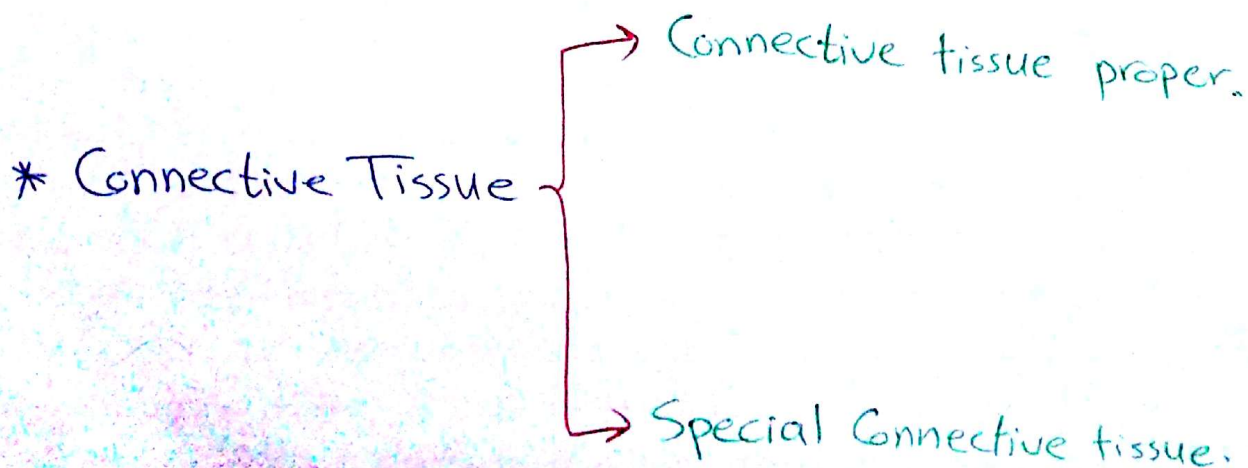
2* Connective Tissue :

- Connective tissues are fibrous tissues made up of cells separated by non-living material called **extracellular matrix**.
- Supports, connects, or separates different types of tissues & organs.
- Found between other tissues everywhere in the body, including the central nervous system.



* **Blood & lymph** lack the fiber components, they are immersed in water

* Cells of Connective tissue → fibroblasts, adipocytes, mast cells, macrophages & leukocytes.



Connective Tissue Proper

Loose (Areolar)

- The most common type of CT in vertebrates.
- Holds organs in place and attaches epithelial tissue to other underlying tissues.
- Surrounds blood vessels and nerves.
- Matrix in this tissue consists of many different cells like mast cells, Microphages & the fibroblasts.
- The matrix contains 3 major types of connective tissue fibers:- Collagenous, Elastic and Reticular fibers.

Composed of collagen fibers with crowded rows of fibroblasts forming a strong rope like structures such as tendons and ligaments.

Dense (CT)

Connective Tissue

Dense irregular CT

Collagen bundles and matrix are distributed in irregular patterns, sometimes it's found in the form of network.

Dense regular CT

Collagen fibers in dense regular CT are bundled in a parallel fashion. In this kind of tissue elastic and reticular fibers are completely absent.

Special Connective Tissue

Reticular

CT

- has a network of reticular fibers that made up of collagen.
- fibers form skeleton (stroma) to support the lymphoid organs (lymph node stromal cells, red bone marrow, and spleen).
- held adipose tissue.

Adipose tissue

(body fat)

- Composed of adipocyte, vascular endothelial cells and adipose tissue macrophages.
- Functions → stores energy in form of lipids / Cushions and insulate the body / Produces hormones (leptin).
- Types:-
 - * White adipose tissue (stores energy)
 - * Brown adipose tissue (generates body heat)

Cartilage

- a flexible CT in animals composed of specialized cells called chondrocytes that produce a large amount of collagenous extracellular matrix (ground substance rich in proteoglycan and elastin fibers).

- classified in 3 types:-

elastic, hyaline, fibrous and all these parts ~~are~~ doesn't contain blood vessels (avascular) or nerves (aneural) so it's supplied by diffusion.

- It's rigid, it often serves the purpose of holding tubes opened in animal's body.

- Found in joints between bones, rib cage, ear, nose, bronchial tubes and intervertebral discs.

Bones

- Consist of living cells embedded in a mineralized (inorganic calcium and phosphate) organic matrix.

- Types:

* Cortical bone: consist of repeated structure called osteon or haversian system. Each osteon has concentric lamellae (layers) of mineralized matrix deposited around ~~central~~ canal called haversian canal contains blood and nerve supply.

* Cancellous bone:

- Bodily fluid in animals that delivers necessary substances such as:-

nutrients and oxygen to the cells & transports metabolic waste products away from those cells.

- Plasma

constitute 55% of blood fluid,

its mostly water (92%)

- Blood Cells:-

① Red Blood Cells (Erythrocytes)

② white BCs (leukocytes)

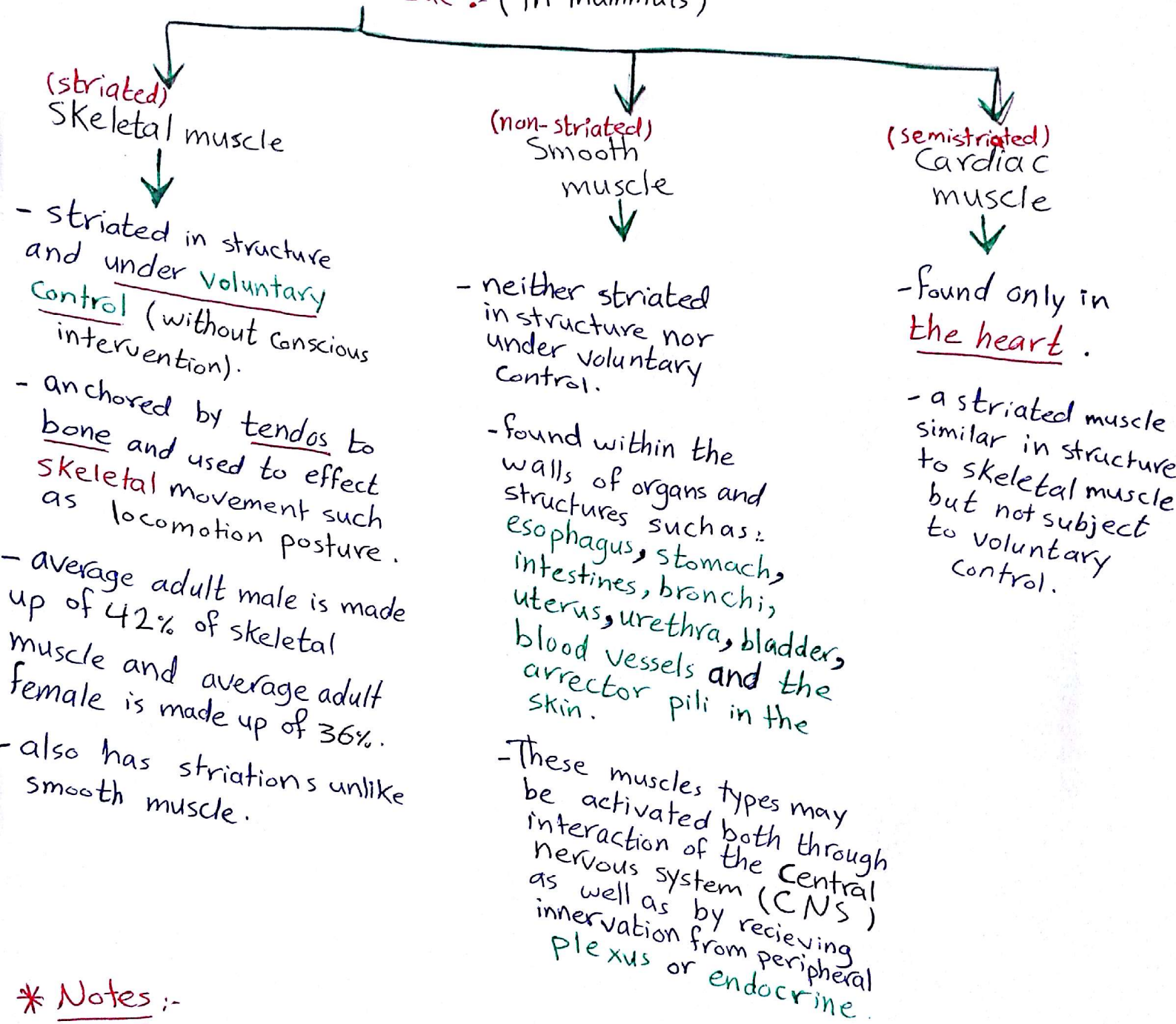
③ Platelets

Blood

3* Muscle Tissue :-

- Soft tissue composed of muscle cells that compose muscles and give them the ability of contraction.

- Types of Muscle Tissue :- (In mammals)



* Notes :-

- * ~~arrector~~ arrector pili → controls erection of body hair.
- * Cardiac & Skeletal muscles are "striated" in that they contain ~~sarcomers~~ sarcomers and packed into highly regular ~~arrang~~ arrangements of bundles; smooth muscles has neither.

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4 * Nervous Tissue :-

- The main component of the two parts of the nervous system; the **brain** and **spinal cord** of the **CNS**, and the branching peripheral nerves of the **peripheral nervous system (PNS)**.
- regulates and controls body activity.
- Composed of **neurons**, or nerve cells, which receive and transmit impulses, and **neuroglia**, known as glial cells which assist propagation of the nerve impulse as well as providing nutrients to the neuron.

* Nervous tissue is categorized into 4 types of tissue. In the CNS, the tissue types ~~are~~ found are:-

1* **Gray matter** → composed of cell bodies, dendrites, ~~un~~ myelinated axons, protoplasmic astrocytes, microglia and very few myelinated axons.

2* **White matter** → composed of myelinated axons, fibrous astrocytes, myelinating oligodendrocytes, and microglia. In the PNS, the tissue types are **nerves** and **ganglia**.

* **Functions of the ~~nervous~~ nervous system:-**

- 1- sensory input.
- 2- integration.
- 3- control of muscles and glands
- 4- homeostasis.
- 5- mental activity.

* Good Luck ^^ *

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(8) * Biotechnology & Genetic Engineering *